Learned representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

Why should a debugging trace inspect scores before ReLU as well as final activations?

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Trace an input through a hidden representation and a decision rule. Try the recall before opening notes.

Your learning target

I can trace an **input** through a hidden **representation** and a decision rule.

I can identify distinctions that a **representation** preserves or loses.

I can separate a useful constructed feature from **evidence** of successful learning.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace an **input** through a hidden **representation** and a decision rule

Identify distinctions that a **representation** preserves or loses

Separate a useful constructed feature from **evidence** of successful learning

Retrieve what you already know

> Why should a debugging trace inspect scores before ReLU as well as final activations?

My starting explanation

This week's end goal

By the end, I can explain learned **representation**s using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Learned representations case. Separate observations, given facts and assumptions before choosing an explanation.

Consider fictional rectangles with **input** [width, height] in cm. The task is to identify rectangles whose width is strictly greater than height. The target is 1 for wider rectangles and 0 otherwise.

A square therefore has target 0. This is an arithmetic teaching task, not a claim that a network is needed to measure rectangles.

Define two hidden units: h1 = ReLU(width - height) and h2 = ReLU(height - width). Their weight rows are [1, -1] and [-1, 1], with zero biases. The **representation** h = [h1, h2] records excess width or excess height.

Predict 1 when h1 > 0; otherwise predict 0.

Input	Hidden representation	Prediction
[4, 2]	[2, 0]	1
[2, 4]	[0, 2]	0
[3, 3]	[0, 0]	0
[5, 3]	[2, 0]	1

These weights were chosen by the lesson author. They demonstrate a **representation** that could be implemented by a network, but no **training** run has established that a model learned it.

Learning would require adjusting parameters using **training** data and recording the procedure.

List the given facts and one fact that is not supplied.

See how it works

> Follow the mechanism for Learned representations. Connect each labelled input, deciding step and result.

A **prediction** head is the computation that reads the hidden vector and produces an **output**. Keeping it separate lets us inspect what information is available before making a decision.

This example assumes two finite, positive measurements in cm; it is not a general **input** validator.

```
def represent(width, height):
    return [max(0, width - height),
            max(0, height - width)]

def predict(hidden):
    return int(hidden[0] > 0)

examples = [(4, 2), (2, 4), (3, 3), (5, 3)]
for width, height in examples:
    hidden = represent(width, height)
    print([width, height], hidden, predict(hidden))

assert represent(4, 2) == represent(5, 3)
assert predict(represent(3, 3)) == 0
```

The raw difference width - height can be recovered as h1 - h2. The original width and height cannot in general be recovered: many rectangles share a difference.

Two hidden components do not guarantee preservation of two **input** components.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Learned representations. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Change one thing in the pictured case

Change one thing

Change: The head keeps both ordered hidden values instead of their sum.

Predicted result

Prediction: It can distinguish [3, 0] from [0, 3] in this example.

Why the result follows

Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Learned representations to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Representation	An encoding of an example used by another computation
Hidden activation	A value produced inside a network after applying an activation function
Collision	Two different inputs receiving the same representation
Learned feature	A feature whose computation depends on parameters adjusted using training data

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Learned representations, predict a step, then compare with its explanation.

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Learned representations

See when a **representation** preserves a distinction and when it destroys it.

The facts we will use

Supplied facts

Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)]. A=[5, 2]; B=[2,

5]; target is left-brighter versus right-brighter.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/19

A representation can preserve direction

	left	right	h1	h2
A	5	2	3	0
B	2	5	0	3

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/19

Learned representations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Encode A

A becomes [3, 0], with left brighter.



2. Encode B

B becomes [0, 3], with right brighter.



3. Compare collapsed summary

If the head receives only h1+h2, both become3 and the target distinction is lost.

CHANGE ONE THING

The head keeps both ordered hidden values instead of their sum.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode A	A becomes [3, 0], with left brighter.
2. Encode B	B becomes [0, 3], with right brighter.
3. Compare collapsed summary	If the head receives only h1+h2, both become3 and the target distinction is lost.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

What this example does not establish

Keep the stated limit

This is a hand-chosen **representation**, not **evidence** that a trained image model actually

learned these units.

Keep the stated limit

Read the labelled picture: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: The head keeps both ordered hidden values instead of their sum. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Learned representations claim and explain a repair using the supplied evidence.

Check the claim against the facts

A **representation** can be sufficient for one task and insufficient for another. Our hidden vector supports the wider-than-height rule for every valid **input** under the stated formula.

It fails to identify exact area for all rectangles because a **collision** can contain different areas. No more complicated deterministic head can recover information that is absent from its **input**.

Compare a second encoding s = width + height. Rectangles [4, 2] and [2, 4] both give s = 6 but have different wider-than-height targets. A head receiving only s must fail on at least one of them.

Useful features depend on the question, not just the number of encoded values.

One repair for area is to retain the original measurements. Another is to append s to h. Since d = h1 - h2, width = (s + d) / 2 and height = (s - d) / 2.

This works with exact valid measurements; it does not mean every added feature improves generalisation on noisy data.

Locate the earliest unsupported step

The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

> **State one repair and the checks you would rerun.**

STUDENT LESSON · 8 OF 18

Practise together

> Practise Learned representations with support, then explain the deciding step in your own words.

Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

Encode [6, 4] and [7, 5]. Give their wider-than-height predictions and areas. Can a head that sees only h always recover the exact area? Explain using this pair.

Explain your prediction to a partner before comparing results.

Practise with less help: Prove an information loss

Use workbook W2 for the connected example “Learned **representation**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Learned **representation**s. Proposed explanation: “Two hidden values preserve both **input** measurements”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**
>
>
>
>
>
>
>
>

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Learned representations. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Give the hidden vectors and areas for [4, 2] and [5, 3]. Explain why increasing only the complexity of a deterministic area head cannot make both exact **prediction**s correct.

```
def represent(width, height):
    return [max(0, width - height),
            max(0, height - width)]

def predict(hidden):
    return int(hidden[0] > 0)

examples = [(4, 2), (2, 4), (3, 3), (5, 3)]
for width, height in examples:
    hidden = represent(width, height)
    print([width, height], hidden, predict(hidden))

assert represent(4, 2) == represent(5, 3)
assert predict(represent(3, 3)) == 0
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Learned representations. Explain why each step is justified.

Explain → apply → challenge

Encode [8, 3], [3, 8] and [4, 4] with the lesson layer. Apply its **prediction** rule and explain the square boundary.

The alternative **representation** is s = width + height. Use [4, 2] and [2, 4] to decide whether s alone always supports the wider-than-height task.

For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled learned **representation**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace an **input** through a hidden **representation** and a decision rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

In a trained network, hidden features depend on fitted weights and may not have simple human names. A strong **output** result does not by itself show that a unit represents a concept such as shape or fairness.

Inspect activations on deliberately varied examples, compare with a baseline, and limit conclusions to the **evidence**.

> **A teammate calls the hand-chosen rectangle layer a learned area detector because all four wider-than-height predictions are correct. Identify two problems with this claim and propose one decisive area check.**

When testing a learned **representation**, fit pre**process**ing and parameters on **training** data, use validation data for choices, and reserve the final test for the declared final evaluation.

Compare **representation**s with matched data, head rules and **training** budgets, and report **collision**s and failures. Next week makes the axis and channel meanings of image tensors explicit.

For h = [2, 0] and s = 8, recover width, height and area. Does this algebra prove that a **training** program learned the **representation** or that it will work well on noisy measurements?

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Learned representations case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Different rectangles can share a hidden vector

INPUT IN CM	HIDDEN VECTOR	AREA IN SQUARE CM
Width 4 height 2	[2, 0]	8
Width 5 height 3	[2, 0]	15

The representation preserves the wider-than-height answer for these examples but discards information needed to recover their different areas. A deterministic head receiving only this vector must give both examples the same output.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Audit a new representation independently

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A different layer uses g = [ReLU(width - height - 1), ReLU(height - width - 1)]. Encode [4, 3] and [3, 4]. State their targets for width strictly greater than height.

Can any deterministic head receiving only g classify both correctly?

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Learned representations, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: See when a **representation** preserves a distinction and when it destroys it.

By the end, I can	How I will show it
Trace an input through a hidden representation and a decision rule	Explain the deciding step, show a trace or test, and state one limit.
Identify distinctions that a representation preserves or loses	Explain the deciding step, show a trace or test, and state one limit.
Separate a useful constructed feature from evidence of successful learning	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Learned representations without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace an **input** through a hidden **representation** and a decision rule

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

See when a **representation** preserves a distinction and when it destroys it.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If the head receives only h1+h2, both become3 and the target distinction is lost.

1. Encode B

2. Encode A

3. Compare collapsed summary

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Compare collapsed summary If the head receives only h1+h2, both become3 and the target distinction is lost.

A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The head keeps both ordered hidden values instead of their sum. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. It can distinguish [3, 0] from [0, 3] in this example.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: It can distinguish [3, 0] from [0, 3] in this example. The **representation** retains the direction of brightness difference when coordinates stay separate.

Repair a missing building block

Review Debugging shapes and gradients (ai-5/18). Goal: Check dimension contracts before computing a layer. Short practice: What should a debugging record pair with every gradient value?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Regularisation and early stopping (ai-5/16). Goal: Separate a weight penalty from **prediction** error. Short practice: Why should the penalty value not be reported as validation error?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

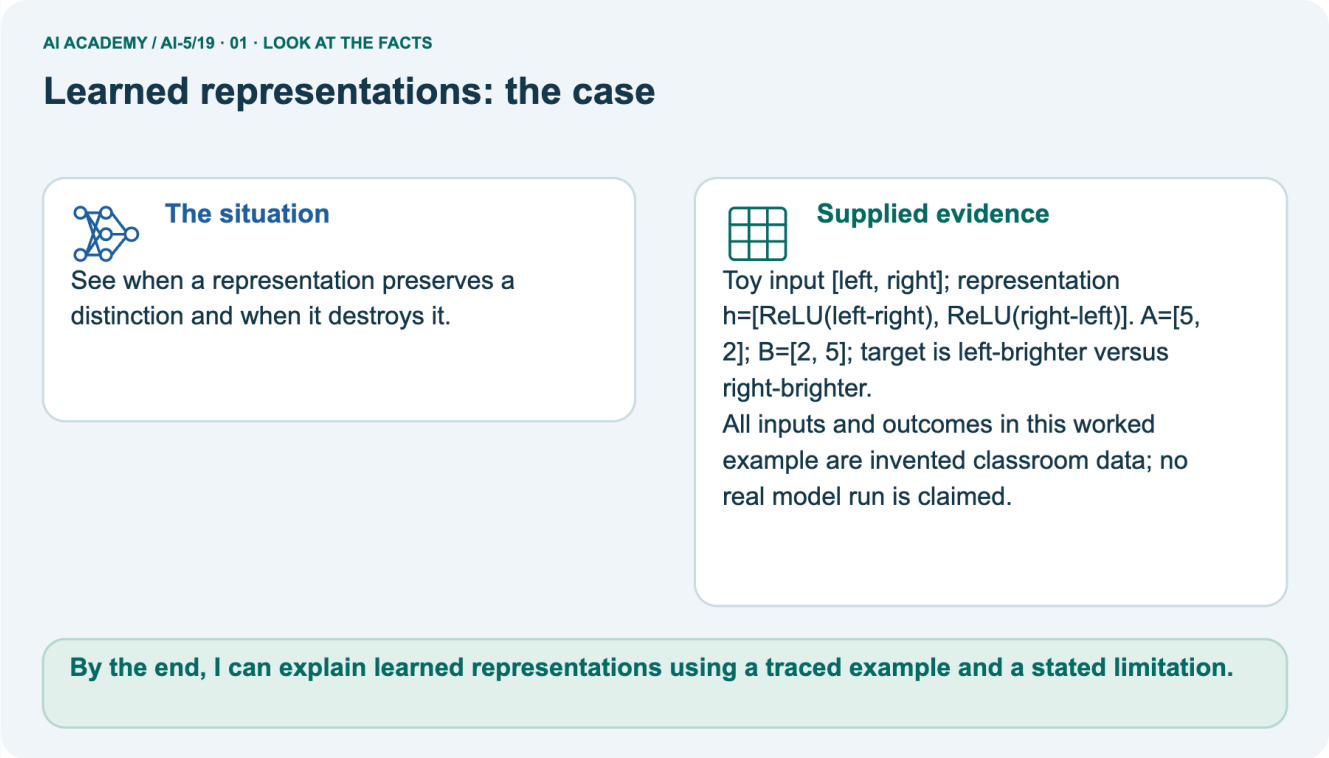
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Learned representations graphic using its labels, deciding evidence and relevant limitation.

01. Learned representations: the case



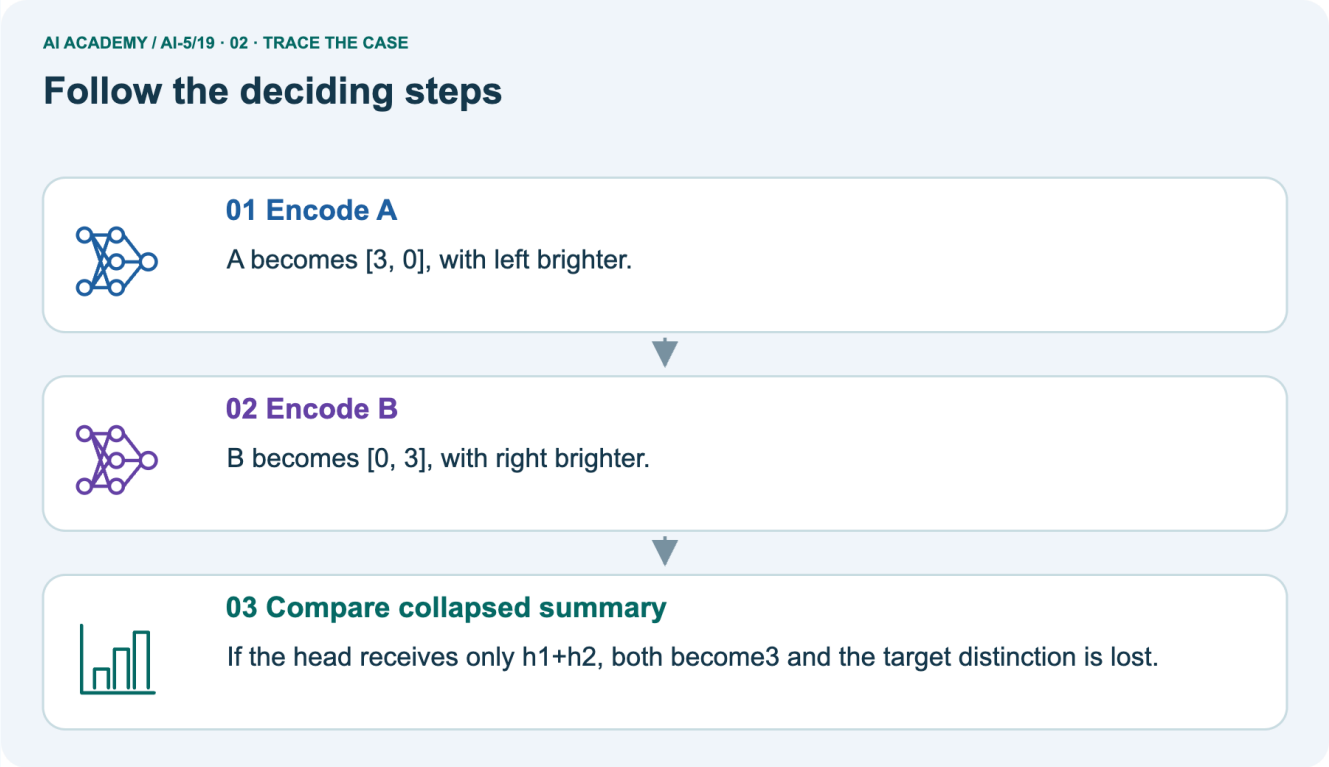
Follow the labelled picture with the matching explanation. Open the vector diagram

See when a **representation** preserves a distinction and when it destroys it. Supplied facts: Toy **input** [left, right]; **representation** h=[ReLU(left-right), ReLU(right-left)].

A=[5, 2]; B=[2, 5]; target is left-brighter versus right-brighter. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

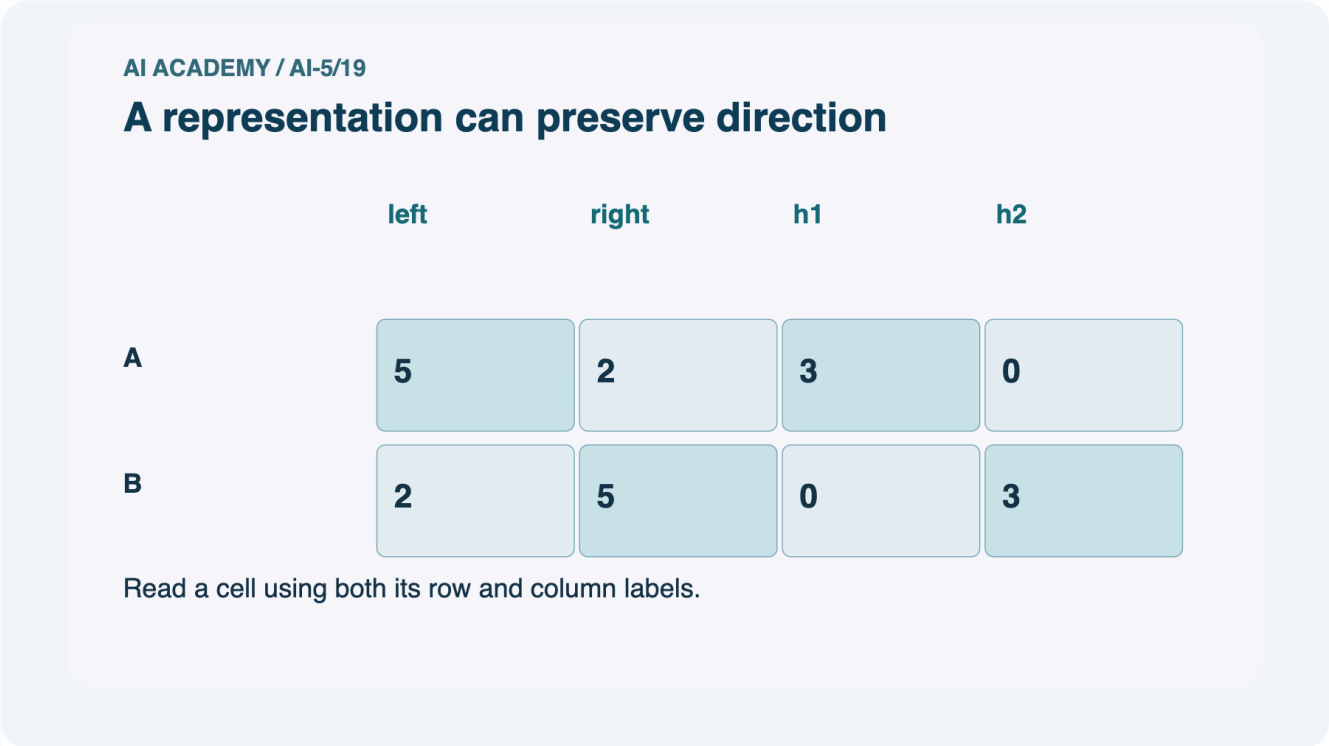


Follow the labelled picture with the matching explanation. Open the vector diagram

Encode A: A becomes [3, 0], with left brighter. → Encode B: B becomes [0, 3], with right brighter. → Compare collapsed summary: If the head receives only h1+h2, both become3 and the target distinction is lost.

Point to each numbered stage and explain which recorded **input** it uses.

03. A representation can preserve direction

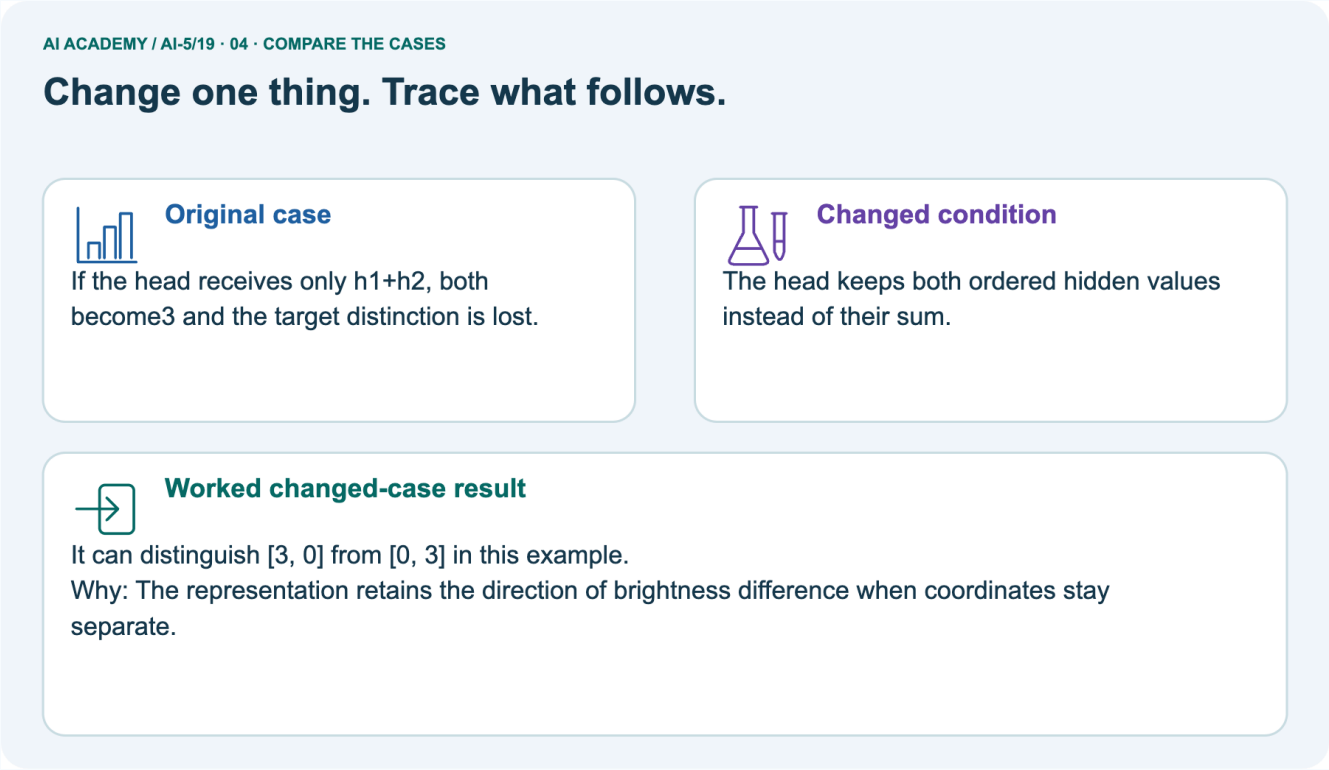


Follow the labelled picture with the matching explanation. Open the vector diagram

A **representation** can preserve direction. A: left = 5, right = 2, h1 = 3, h2 = 0. B: left = 2, right = 5, h1 = 0, h2 = 3.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



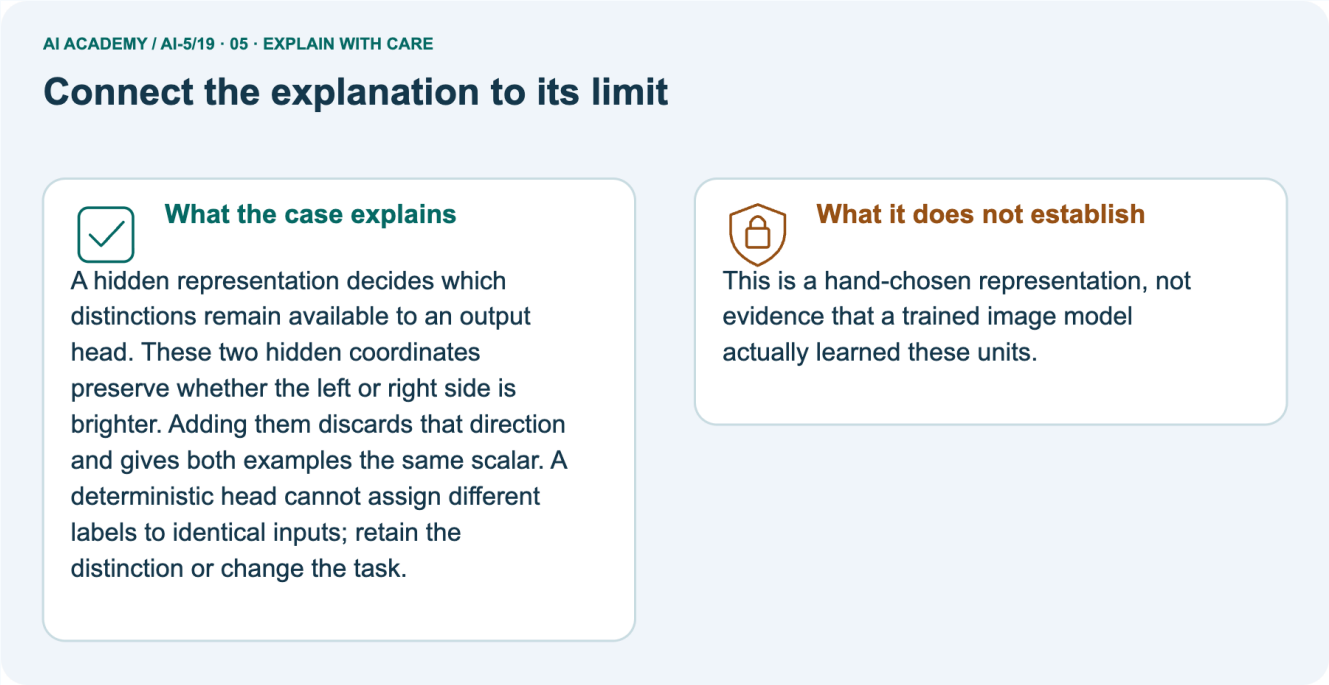
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If the head receives only h1+h2, both become3 and the target distinction is lost. Changed condition: The head keeps both ordered hidden values instead of their sum.

Expected result: It can distinguish [3, 0] from [0, 3] in this example. Reason: The **representation** retains the direction of brightness difference when coordinates stay separate.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

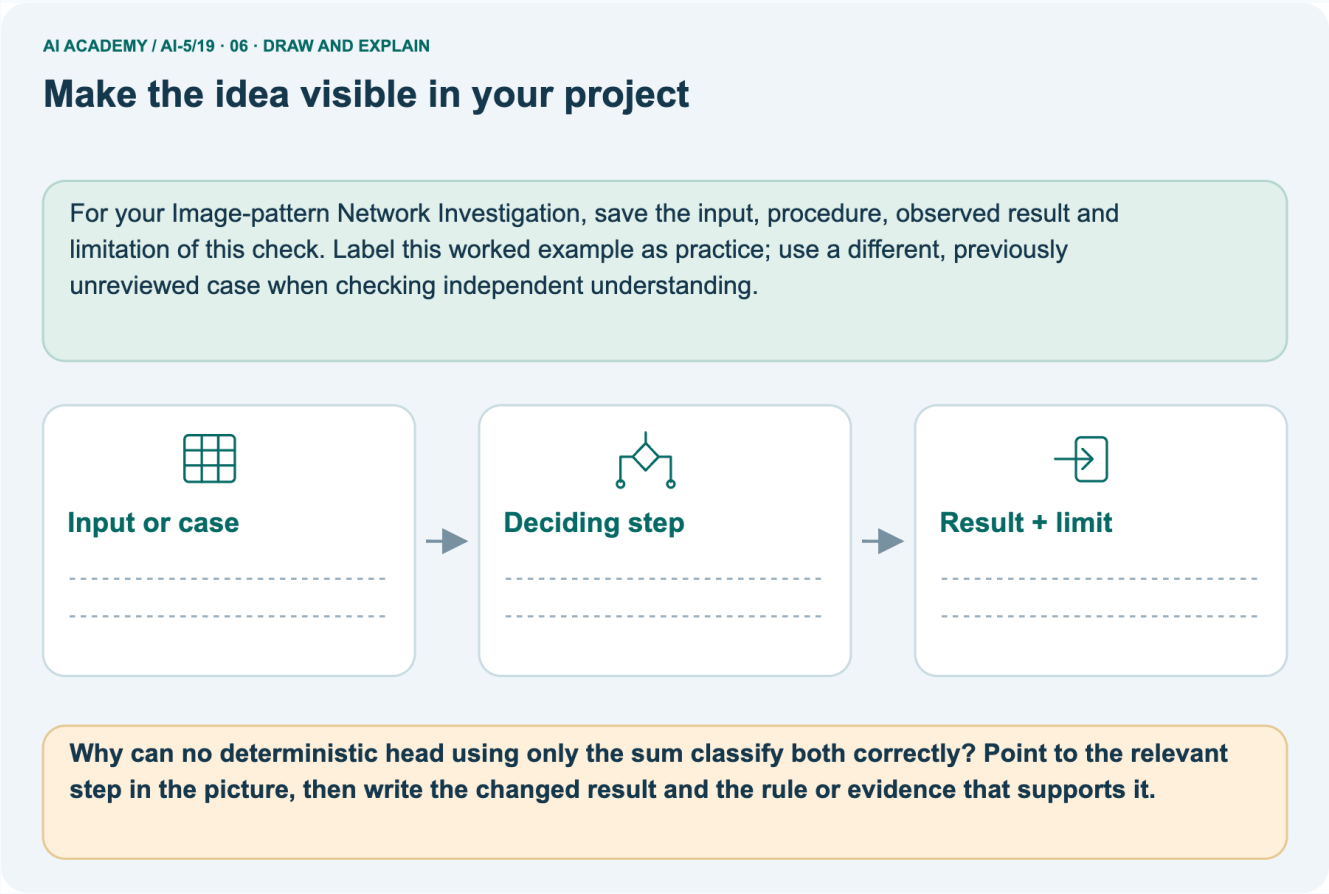
A hidden **representation** decides which distinctions remain available to an **output** head. These two hidden coordinates preserve whether the left or right side is brighter.

Adding them discards that direction and gives both examples the same scalar. A deterministic head cannot assign different labels to identical **input**s; retain the distinction or change the task.

Limit: This is a hand-chosen **representation**, not **evidence** that a trained image model actually learned these units.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why can no deterministic head using only the sum classify both correctly? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.